

Ex. No. : 7

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NAME:SHAMRUDHA VARSHINI.P

ROLLNO:231901048

IPC USING SHARED MEMORY

Aim:

To write a C program to implement Inter Process Communication (IPC) using shared memory between sender and receiver processes.

Algorithm:

Sender Process

1. Set the size of the shared memory segment.
2. Allocate the shared memory segment using shmget().
3. Attach the shared memory segment using shmat().
4. Write a string to the shared memory segment using sprintf().
5. Set delay using sleep().
6. Detach shared memory segment using shmdt().

Receiver Process

1. Set the size of the shared memory segment.
2. Allocate the shared memory segment using shmget().
3. Attach the shared memory segment using shmat().
4. Print the shared memory contents sent by the sender process.
5. Detach shared memory segment using shmdt().

Program Code:

sender.c

```
#include <stdio.h>
#include <sys/ipc.h>
#include <sys/shm.h>
#include <unistd.h>
#include <string.h>
```

```

int main() {    key_t key = ftok("shmfile",65); // Generate unique key    int
shmidx = shmget(key, 1024, 0666|IPC_CREAT); // Create shared memory
char *str = (char*) shmat(shmidx, (void*)0, 0); // Attach to shared memory

    sprintf(str, "Welcome to Shared Memory");
printf("Message Sent: %s\n", str);

    sleep(5); // Delay to allow receiver to read
shmidx(shmidx); // Detach from shared memory

    return 0;
}

```

receiver.c

```

#include <stdio.h>
#include <sys/ipc.h>
#include <sys/shm.h>
#include <unistd.h>

int main() {    key_t key = ftok("shmfile",65); // Generate same key    int
shmidx = shmget(key, 1024, 0666|IPC_CREAT); // Access shared memory
char *str = (char*) shmat(shmidx, (void*)0, 0); // Attach to shared memory

    printf("Message Received: %s\n", str);

    shmidx(shmidx); // Detach from shared memory    shmctl(shmidx,
IPC_RMID, NULL); // Destroy the shared memory    return 0;

}

```

Sample Output:

Terminal 1:

```
[root@localhost student]# gcc sender.c -o sender
```

```
[root@localhost student]# ./sender Message
```

Sent: Welcome to Shared Memory

Terminal 2:

```
[root@localhost student]# gcc receiver.c -o receiver
```

```
[root@localhost student]# ./receiver
```

Message Received: Welcome to Shared Memory

Result:

Thus, the C program for Inter Process Communication (IPC) using shared memory was successfully executed, and the message was successfully passed from the sender process to the receiver process.